Cisco 642-785

Maintaining Cisco Service Provider Quality of Service

Version: 5.1

QUESTION NO: 1

Within a QoS policy-map configuration, one of the traffic classes is configured with the random-detect dscp-based configuration command. What will that command accomplish?

A. provides congestion management by queueing the traffic using CBWFQ
B. provides congestion management by queueing the traffic using LLQ
C. decreases the probability of congestion by selectively dropping TCP packets before the queue is full
D. provides congestion avoidance by selectively delaying the delivery of packets with lower DSCP priority
E. decreases congestion by avoiding global UDP synchronization

Answer: C
Explanation:

QUESTION NO: 2

What are three benefits of IntServ and RSVP? (Choose three.)

A. RSVP helps network devices identify dynamic port numbers.
B. IntServ networks will reject or downgrade new RSVP sessions if all reservable bandwidth is booked somewhere in a path.
C. RSVP signaling is a scalable way to ensure all devices maintain an accurate picture of the network state.
D. They enable the network to guarantee necessary QoS to individual data flows.
E. The IntServ class-based approach is easy to design and implement.

Answer: A,B,D
Explanation:

QUESTION NO: 3

Which statement about the rates and statistics (such as match, transmit, drop, and police) shown in the show policy-map interface output is true?

A. The rates are computed as a moving average of instantaneous rates.
B. The rates are displayed in real time, which matches the actual traffic rate.
C. The matched statistics of the packet can be cleared only when the router reboots.
D. The matched statistics of the packet are displayed in real time.
E. The traffic policing statistics are displayed in real time.

Answer: A

Explanation:

QUESTION NO: 4

Which four factors can be used for packet classification in a QoS-aware network device? (Choose four.)

A. source address
B. destination address
C. DSCP
D. TTL
E. MQC
F. IP precedence

Answer: A,B,C,F

Explanation:

QUESTION NO: 5

What are the three primary types of faults associated with QoS fault management? (Choose three.)

A. classification faults
B. buffer faults
C. queue faults
D. marking faults
E. assurance faults
F. physical faults

Answer: A,D,E

Explanation:

QUESTION NO: 6

Which are the two locations where the SP network device always trusts the QoS markings from its upstream neighbor? (Choose two.)

A. at the ingress PE
B. at the egress PE
C. in the SP core
D. at the ingress CE
E. at the egress CE

Answer: B,C
Explanation:

QUESTION NO: 7

Which of these correctly describes traffic classification using qos-group?

A. qos-group marking is automatically mapped to MPLS EXP marking.
B. qos-group is only applicable to an MPLS-enabled router.
C. qos-group marking value ranges from 0 to 7.
D. qos-group is local to the router.

Answer: D
Explanation:

QUESTION NO: 8

Which two IP SLA Probe types can be used to measure voice quality? (Choose two.)

A. HTTP
B. ICMP Path Jitter
C. UDP Echo
D. UDP Jitter
E. UDP Delay
F. UDP MOS

Answer: B,D
Explanation:

QUESTION NO: 9

Refer to the exhibit.

```
class-map match-any REALTIME
  match dscp ipv4 46
  match dscp ipv4 40
!
class-map match-all CLASS-A
  match protocol SAP
!
class-map match-any CLASS-B
  match dscp ipv4 10
!
policy-map POLICY-SHAPE
  class REALTIME
   bandwidth percent 25
  class CLASS-A
   bandwidth percent 20
  class CLASS-B
      bandwidth 192
      shape average 128000
       queue-limit 64
```

Traffic in CLASS-B is not getting a minimum bandwidth of 192 kb/s in this policy map. What can be done to correct this?

A. No changes are needed, the site is already correct.
B. The shape peak command should be used instead of the shape average command.
C. Set the shape rate to a CIR higher than 192 kb/s.
D. Decrease the maximum queue size.

Answer: C
Explanation:

QUESTION NO: 10

Which of these describes Short Pipe mode in QoS for MPLS VPN service providers?

A. Service provider can remark customer DSCP values.
B. Service provider does not remark customer DSCP values. (Service provider uses independent MPLS EXP markings.) Final PE-to-CE policies are based on service provider markings.
C. The customer and service provider share the same DiffServ domain.
D. Service provider does not remark customer DSCP values. (Service provider uses independent MPLS EXP markings.) Final PE-to-CE policies are based on customer markings.

Answer: D
Explanation:

QUESTION NO: 11

Refer to the exhibit.

```
Router(config)# policy-map cust1-classes
Router(config-pmap)# class gold
Router(config-pmap-c)# bandwidth percent 50
Router(config-pmap)# class silver
Router(config-pmap-c)# bandwidth percent 20
Router(config-pmap)# class bronze
Router(config-pmap-c)# bandwidth percent 15
!
Router(config)# policy-map cust2-classes
Router(config-pmap)# class gold
Router(config-pmap-c)# bandwidth percent 30
Router(config-pmap)# class silver
Router(config-pmap-c)# bandwidth percent 15
Router(config-pmap)# class bronze
Router(config-pmap-c)# bandwidth percent 10
!
Router(config)# policy-map cust-policy
Router(config-pmap)# class cust1
Router(config-pmap-c)# shape average 384000
Router(config-pmap-c)# service-policy cust1-classes
Router(config-pmap)# class cust2
Router(config-pmap-c)# shape peak 51200
Router(config-pmap-c)# service-policy cust2-classes
!
Router(config-pmap-c)# interface Serial 3/2
Router(config-if)# service out cust-policy
```

Which two statements are correct? (Choose two.)

A. The policy map cust-policy specifies an average traffic shaping rate of 384 kb/s for the cust1 class and assigns the service policy cust1-classes to the cust1 class.
B. This example is a hierarchical MQC example, and the cust-policy is the child policy.
C. This example queues the traffic into different CBWFQ classes before the aggregate traffic is shaped.
D. Within the cust1-classes policy, the gold class is guaranteed 50 percent of the bandwidth. Silver is guaranteed 20 percent of the bandwidth, and bronze is guaranteed 15 percent of the bandwidth.
E. The traffic class cust2 is ensured a bandwidth of 51.2 kb/s.
F. Within the cust2-classes policy, the gold class is guaranteed 30 percent of the S3/2 interface bandwidth. Silver is guaranteed 15 percent of the S3/2 interface bandwidth. Bronze is guaranteed 10 percent of the S3/2 interface bandwidth.

Answer: A,D
Explanation:

QUESTION NO: 12

Refer to the exhibit.

```
policy-map Customer1
  class mission-critical
  bandwidth 256
  shape average 192000
  queue-limit 64
```

A customer's policy profile was updated to increase the minimum bandwidth to 256 kb/s for mission critical data, but reports indicate that the policy now has an error. What change should be made to the policy to correct this error?

A. The bandwidth statement should read 256000.
B. Set the shape rate to a CIR that is higher than 256 kb/s.
C. Use the shape peak command instead of the shape average command.
D. Increase the maximum queue size to 128 to compensate for the mission critical data.
E. The priority command should have been used, not the bandwidth command.

Answer: B

Explanation:

QUESTION NO: 13

Which two statements about IP SLA are correct? (Choose two.)

A. Cisco IOS IP SLA is Layer 2 transport independent.
B. IOS IP SLA uses active traffic monitoring which allows users to analyze service levels for IP applications.
C. IOS IP SLA is a traffic monitoring mechanism that gathers statistics about traffic based on packet headers and fields.
D. Cisco IOS IP SLA can be used to help detail billing information for each defined class of service.
E. IP SLA uses NBAR to provide the reporting engine for statistics that it collects.

Answer: A,B

Explanation:

QUESTION NO: 14

You have deployed IP SLA in your network to monitor VoIP. You want to be notified on network delay before it affects voice quality. What threshold should be set?

A. 150 ms one-way
B. 150 ms round trip
C. 50 ms round trip
D. 250 ms one-way
E. 200 ms round trip

Answer: A

Explanation:

QUESTION NO: 15

When troubleshooting a QoS problem, which two key statistics should you look for? (Choose two.)

A. queue utilization

B. interface utilization
C. queue priority
D. queue drops
E. queue depth

Answer: D,E
Explanation:

QUESTION NO: 16

Which three NetFlow versions are supported by Cisco NetFlow Analyzer? (Choose three.)

A. 1
B. 5
C. 6
D. 7
E. 8
F. 9

Answer: A,B,F
Explanation:

QUESTION NO: 17

Which command shows real traffic statistics on each class map on the 10 Gigabit interface 0/10/0/0 at the egress direction?

A. show service-policy interface tenGigE 0/10/0/0 output
B. show service-policy interface tenGigE 0/10/0/0 egress
C. show policy-map interface tenGigE 0/10/0/0 output
D. show policy-map interface tenGigE 0/10/0/0 egress
E. show qos interface tenGigE 0/10/0/0 output
F. show qos interface tenGigE 0/10/0/0 egress

Answer: C
Explanation:

QUESTION NO: 18

When is a best-effort QoS policy most appropriate?

A. when the customer has high bandwidth access to the network
B. when the customer will only transport delay-tolerant data
C. when the CE is self-managed and the customer provides adequate QoS marking
D. when the customer plans to transport data from no more than one multimedia application

Answer: B
Explanation:

QUESTION NO: 19

Which Cisco IOS command will show a summary of probes?

A. show ip sla statistics
B. show ip sla responder
C. show ip sla monitors
D. show ip sla configuration

Answer: D
Explanation:

QUESTION NO: 20

Where do service providers apply policy when traffic is leaving the customer network? (Choose two.)

A. Apply input policy on the CE interface toward PE, if CE is unmanaged.
B. Apply input policy on the PE interface toward CE, if CE is unmanaged.
C. Apply input policy on the PE interface toward CE, if CE is managed.
D. Apply output policy on the PE interface toward CE, if CE is managed.
E. Apply output policy on the CE interface toward PE, if CE is managed.

Answer: B,E
Explanation:

QUESTION NO: 21

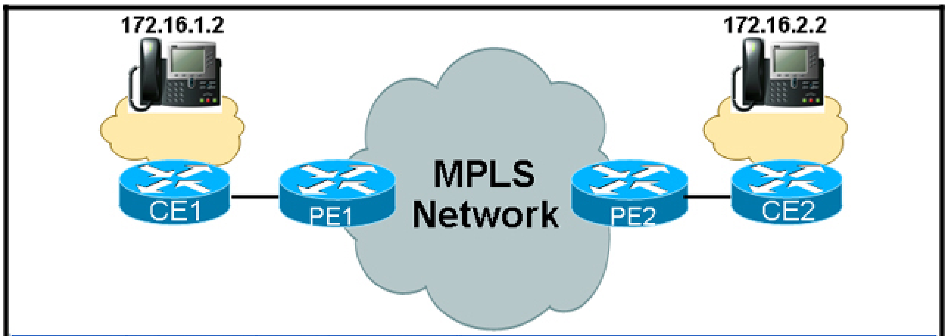
Which two methods can be used to monitor QoS performance? (Choose two.)

A. SNMP
B. SYSLOG
C. TFTP
D. RMON
E. NetFlow

Answer: A,E
Explanation:

QUESTION NO: 22

Refer to the exhibit.




```
⊞ Frame 42 (78 bytes on wire, 78 bytes captured)
⊟ Ethernet II, Src: Cisco_ff:a0:c1 (00:07:0e:ff:a0:c1), Dst: Cisco_d5:b9:80 (00:07:0e:d5:b9:80)
  ⊞ Destination: Cisco_d5:b9:80 (00:07:0e:d5:b9:80)
  ⊞ Source: Cisco_ff:a0:c1 (00:07:0e:ff:a0:c1)
    Type: IP (0x0800)
⊟ Internet Protocol, Src: 172.16.1.2 (172.16.1.2), Dst: 172.16.2.2 (172.16.2.2)
    Version: 4
    Header length: 20 bytes
  ⊞ Differentiated Services Field: 0xa0 (DSCP 0x28)
    Total Length: 64
    Identification: 0x2020 (8224)
  ⊞ Flags: 0x00
    Fragment offset: 0
    Time to live: 126
    Protocol: UDP (0x11)
  ⊞ Header checksum: 0xc0c8 [correct]
    Source: 172.16.1.2 (172.16.1.2)
    Destination: 172.16.2.2 (172.16.2.2)
⊟ User Datagram Protocol, Src Port: 49594 (49594), Dst Port: 49606 (49606)
    Source port: 49594 (49594)
    Destination port: 49606 (49606)
    Length: 44
  ⊞ Checksum: 0x13b6 [validation disabled]
⊞ Data (36 bytes)

0000  00 07 0e d5 b9 80 00 07  0e ff a0 c1 08 00 45 a0   ........ .......E.
0010  00 40 20 20 00 00 7e 11  c0 c8 ac 10 01 02 ac 10   .@  ..~. ........
0020  02 02 c1 ba c1 c6 00 2c  13 b6 80 04 b6 a5 00 09   ......., ........
0030  b0 28 68 74 94 d4 70 b3  7e d5 aa 75 d4 28 79 92   .(ht..p. ~..u.(y.
0040  87 c3 ce 9f a7 f1 42 a9  59 15 30 8c 76 bb         ......B. Y.0.v.
```

```
PE2#show policy-map interface ethernet0/0/1
 Ethernet0/0/1

 Service-policy output: voice_priority

   Class-map: voice (match-all)
     0 packets, 0 bytes
     5 minute offered rate 0 bps, drop rate 0 bps
     Match: ip dscp ef
     Queueing
       Strict Priority
       Output Queue: Conversation 264
       Bandwidth 320 (kbps) Burst 8000 (Bytes)
       (pkts matched/bytes matched) 0/0
       (total drops/bytes drops) 0/0

   Class-map: video (match-all)
     0 packets, 0 bytes
     5 minute offered rate 0 bps, drop rate 0 bps
     Match: ip dscp af41
     Queueing
       Output Queue: Conversation 265
       Bandwidth 50 (%) Max Threshold 64 (packets)
       (pkts matched/bytes matched) 0/0
       (depth/total drops/no-buffer drops) 0/0/0

   Class-map: critical (match-all)
     6647 packets, 757758 bytes
     5 minute offered rate 21000 bps, drop rate 0 bps
     Match: ip dscp cs3
     Queueing
       Output Queue: Conversation 266
       Bandwidth 35 (%) Max Threshold 64 (packets)
       (pkts matched/bytes matched) 1/114
       (depth/total drops/no-buffer drops) 0/0/0

   Class-map: class-default (match-any)
     10729 packets, 1144263 bytes
     5 minute offered rate 6000 bps, drop rate 0 bps
     Match: any
     Queueing
       Output Queue: Conversation 267
       Bandwidth 15 (%) Max Threshold 64 (packets)
       (pkts matched/bytes matched) 15/4085
       (depth/total drops/no-buffer drops) 0/0/0
PE2#
```

Voice quality is poor on calls placed between the two sites. What change would you recommend to solve this problem?

A. Set up WRED to help avoid link congestion.
B. Increase the bandwidth of the voice priority queue.
C. Change the voice class map to match DSCP EF or CS5.
D. Lower the total bandwidth allocated for the video, critical, and class-default queues, because they are starving the voice queue.

Answer: C
Explanation:

QUESTION NO: 23

Which two items can be obtained from the CISCO-NETFLOW-MIB? (Choose two.)

A. buffer hits
B. CEF tables
C. signal-to-noise ratio
D. cache configuration
E. protocol statistics

Answer: D,E
Explanation:

QUESTION NO: 24

Where would you apply WRED to avoid congestion and poor QoS from oversubscription of the network?

A. wherever there is a potential bottleneck
B. wherever there is a potential bottleneck and most traffic is TCP
C. wherever there is a potential bottleneck and traffic is delay-intolerant
D. wherever there is a potential bottleneck and most traffic is UDP
E. wherever there is a potential bottleneck and most traffic is voice

Answer: B
Explanation:

QUESTION NO: 25

Which MQC capabilities are unique to Cisco IOS XR Software?

A. classification and marking
B. congestion management
C. congestion avoidance
D. policing and shaping
E. fabric qos

Answer: E
Explanation:

QUESTION NO: 26

An MPLS VPN customer reports that their DSCP markings from one CE to another CE are being remarked by the service provider. What can cause customer packets to be remarked?

A. The MPLS VPN service provider is using Penultimate Hop Popping.
B. The service provider is using Uniform mode as the MPLS QoS tunnel mode.
C. The service provider is using the Short Pipe mode as the MPLS QoS tunnel mode.
D. The service provider is using Pipe mode as the MPLS QoS tunnel mode.

Answer: B
Explanation:

QUESTION NO: 27

A customer reports that the packets from their custom TCP application, which is currently marked as DSCP AF43, is being dropped more frequently than desired. All of the links in service provider network are using DSCP-based WRED. What can be changed to best improve the drop rate of the customer's custom application without causing major problems for the other traffic classes?

A. Remove WRED on all the links.
B. Mark the custom application traffic to DSCP EF instead.
C. Mark the custom application traffic to DSCP AF41 instead.
D. Use a different queuing method for the AF43 traffic class.
E. Allocate more bandwidth to the AF43 traffic class.

Answer: C
Explanation:

QUESTION NO: 28

Which Cisco IOS XR command will help troubleshoot why a policy map fails to apply on an interface?

A. show service-policy interface tenGigE 0/0/0/0
B. show policy-map interface tenGigE 0/0/0/0
C. show policymgr qos trace all
D. show qos interface tenGigE 0/0/0/0
E. clear qos counters

Answer: C
Explanation:

QUESTION NO: 29

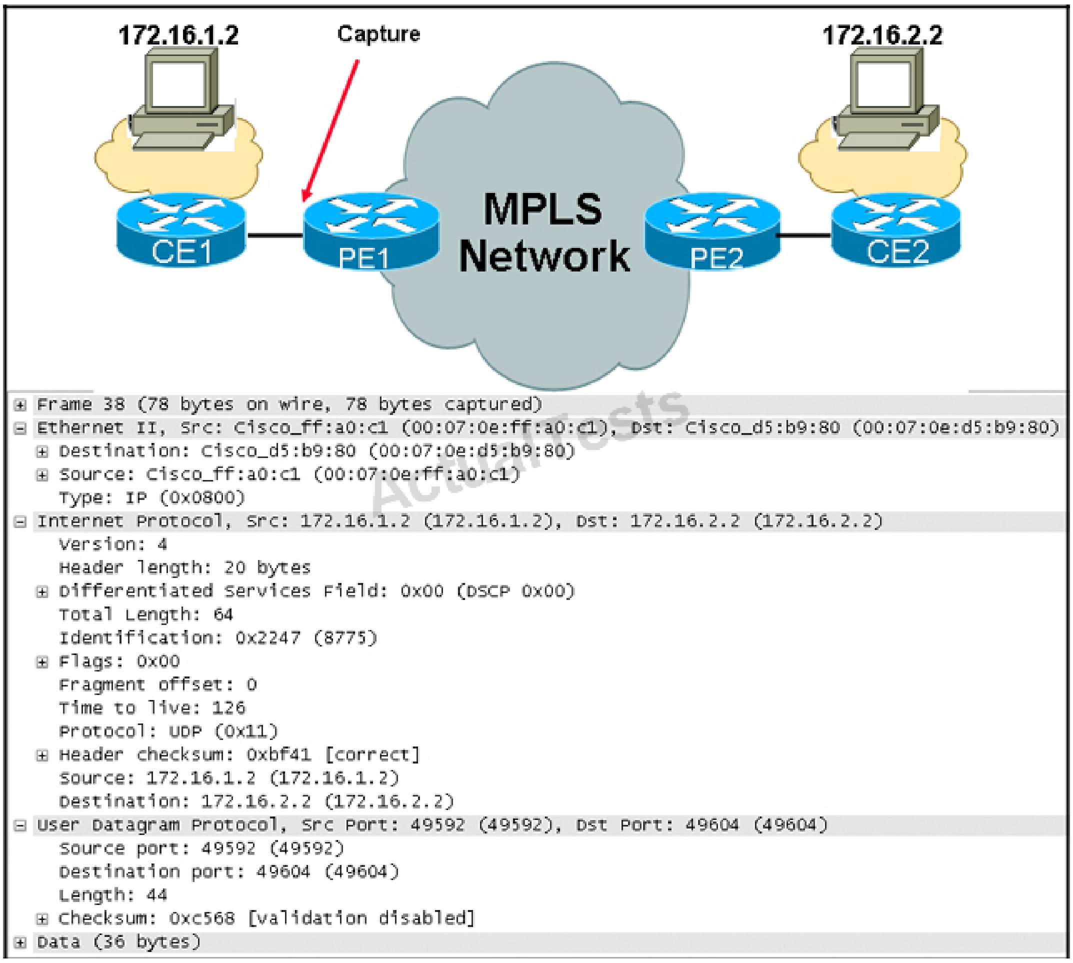
Which Cisco IOS command lists all traffic classifications?

A. show class-map
B. show policy-map
C. show service-policy
D. show qos

Answer: A
Explanation:

QUESTION NO: 30

Refer to the exhibit.



Your MPLS network is using RFC 3270 Uniform mode. Your customer reports that when a Microsoft NetMeeting call is placed between the workstations, the user at 172.16.1.2 experiences acceptable audio quality but the user at 172.16.2.2 experiences poor audio quality. What course of action should you take?

A. Change your policy to not remark audio packets within your network.
B. Have the customer set the correct marking of audio packets at the 172.16.1.2 device.
C. Have the customer set the correct marking of audio packets at the 172.16.2.2 device.
D. Use a policy map at the ingress to the PE1 router to remark packets from 172.16.1.2.
E. Use a policy map at the ingress to the PE2 router to remark packets from 172.16.2.2.

Answer: B

Explanation:

QUESTION NO: 31

Which two tools can be used to monitor and report on if a QoS queue is working on a network device? (Choose two.)

A. CiscoWorks IPM
B. Remedy
C. InfoVista
D. NetCool
E. IP SLA
F. Splunk

Answer: A,C
Explanation:

QUESTION NO: 32

What are the benefits of using differentiated services in the service provider backbone? (Choose two.)

A. simple, easy to design, deploy, and operate
B. less bandwidth required to achieve the same SLA
C. more aggregate traffic supported
D. overprovision by two times the maximum aggregate

Answer: B,C
Explanation:

QUESTION NO: 33

Which statement about the MQC configuration shown here for the class-default traffic class is true?

policy-map CE-EDGE-4CLASS-OC3-POS-1

class class-default

bandwidth percent 25

!

A. The maximum number of packets the queue for the class-default class can hold is 64 packets using the default queue-limit.
B. The bandwidth percent is referenced to the reserved bandwidth on the interface.
C. This configuration is invalid, because the class-default traffic class does not support the bandwidth command.

D. The class-default class will use WFQ instead of FIFO queueing.

Answer: A
Explanation:

QUESTION NO: 34

A customer reports drops for traffic in the BIZ-CRITICAL class. As part of the verification process you identify that there is a high number of random drops. Which of these can help address this problem?

A. Decrease the queue limit for the traffic class.
B. Increase the WRED min threshold value for the traffic class.
C. Enable fair queuing for the BIZ-CRITICAL class.
D. Decrease the WRED drop probability denominator for the class.

Answer: B
Explanation:

QUESTION NO: 35

What three statements about WRED are true? (Choose three.)

A. WRED is a congestion avoidance technique.
B. Using WRED on the edge devices (CE, PE) is an effective method of implementing congestion avoidance in the service provider network, because it avoids the congestion prior to reaching the core.
C. WRED should always be used in traffic classes that support a MIX of realtime UDP and TCP traffic.
D. WRED employs a weighted method for determining which packets to drop to avoid congestion. Packets with a higher precedence are less likely to be dropped.
E. Using WRED will necessitate a large number of dropped packets.

Answer: A,B,D
Explanation:

QUESTION NO: 36

An MPLS VPN customer with managed CE services recently made some changes to their call manager configurations. Now the customer reports that their VoIP traffic voice quality is very poor. What could be a cause of the problem?

A. The VoIP traffic class is not being policed at the CE router egress.
B. The VoIP traffic is being marked as DSCP EF.
C. The customer changed the codec used for voice compression from G.729 to G.711.
D. RTP header compression is used for the VoIP traffic class.
E. The traffic shaping rate for the VoIP traffic class is too low.

Answer: C
Explanation:

QUESTION NO: 37

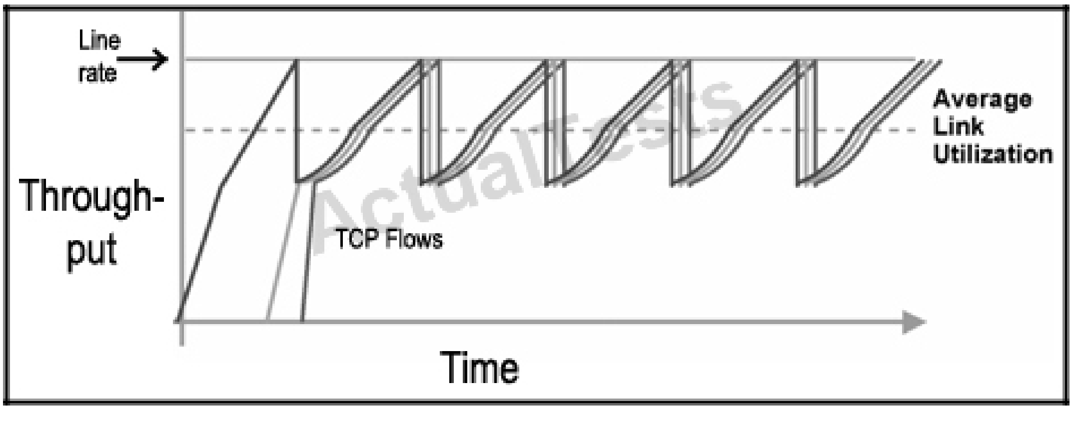
A MPLS VPN customer reports poor VoIP call quality across the MPLS VPN. What could be the problem?

A. All the core links in the service provider network are using LLQ.
B. The service provider is using RTP header compression on the high-speed core links.
C. The total end-to-end delay measured is constantly over 350 msec.
D. The service provider is using the Short Pipe mode rather than the Uniform mode.
E. The service provider has assigned more than 33% of the interface bandwidth to the priority class.

Answer: C
Explanation:

QUESTION NO: 38

Refer to the exhibit.



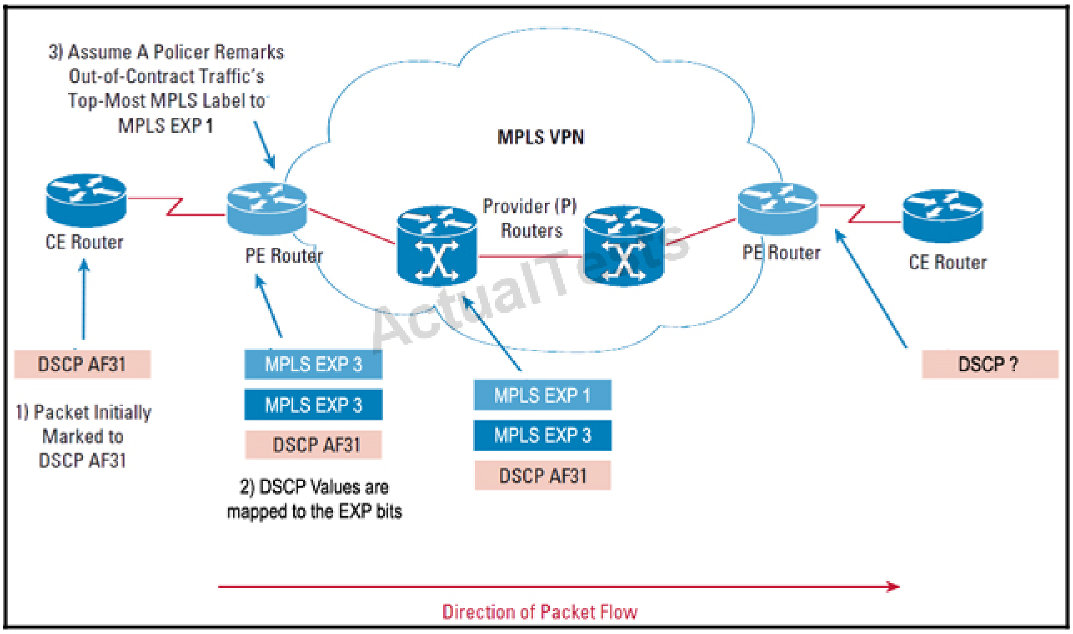
Some customers report that at peak times during the day their average throughput seems to be less than what they expect. While troubleshooting, you notice that data flows sometimes seem to be synchronized with a tendency to increase then drop off in a repeating manner. Which action would help to alleviate this situation?

A. Implement traffic shaping to smooth out the peaks and troughs.
B. Implement WRED to help prevent tail drops during periods of congestion.
C. Change your QoS model from DiffServ to IntServ for better QoS scalability.
D. Decrease the size of output queues in the network to allow the scheduler to allocate link bandwidth fairly.

Answer: B
Explanation:

QUESTION NO: 39

Refer to the exhibit.



You have implemented RFC 3270 Uniform mode in your MPLS network. A customer packet with the DSCP value of AF31 has been sent to the ingress PE router. The PE router has mapped the DSCP values into the MPLS experimental bits, but the policer has remarked the experimental bits of the top label. What will the DSCP value of the packet be when it leaves the egress PE router?

A. BE
B. CS1
C. CS3

D. AF31

Answer: B
Explanation:

QUESTION NO: 40

Which Cisco IOS XR Software command clears the service-policy counters on the 10 Gigabit interface 0/0/0/0 at the ingress direction?

A. clear qos counters tenGigE 0/0/0/0 input
B. clear service-policy counters tenGigE 0/0/0/0 input
C. clear qos counters tenGigE 0/0/0/0 ingress
D. clear service-policy counters tenGigE 0/0/0/0 ingress
E. clear counters qos tenGigE 0/0/0/0 input
F. clear counters service-policy tenGigE 0/0/0/0 input

Answer: A
Explanation:

QUESTION NO: 41

What problem could occur when applying this policy map on CE?

policy-map CE-SP-MODEL

class VOIP

priority

class BROADCAST-VIDEO

priority

class SP-CRITICAL-DATA

bandwidth percent 40

random-detect exp 3 15000 packets 45000 packets class class-default

random-detect exp 1 2500 packets 7500 packets random-detect exp 0 1500 packets 10500 packets

A. All traffic is classified and treated properly, no issues occur.
B. Only one priority queue is allowed per policy.
C. No bandwidth or policing is configured for voice and video traffic, and other traffic could be impacted.
D. Default class does not have bandwidth assigned.

Answer: C
Explanation:

QUESTION NO: 42

Refer to the exhibit.

```
class-map match-any REALTIME
  match dscp ipv4 46
  match dscp ipv4 34
  match mpls experimental topmost 5
!
class-map match-any CRITICAL-DATA
  match dscp ipv4 26
  match mpls experimental topmost 3
!
policy-map PE-OUT
   class REALTIME
      police rate percent 35
          conform-action transmit
          exceed-action drop
        priority
    class CRITICAL-DATA
       bandwidth percent 40
    class class-default
       random-detect exp 1 2500 packets 7500 packets
       random-detect exp 0 1500 packets 10500 packets
!
interface TenGigE0/0/1/1
   service-policy output PE-OUT
```

CLASS	DSCP	EXP
Reserved for Control Plan Traffic	Class Selector 7	7
Reserved for Control Plan Traffic	Class Selector 6	6
Class 1 (real-time traffic)	EF	5
Class 2 in-profile	AF31	4
Class 2 out-of-profile	AF32,AF33	3
Class 3 in-profile	AF11	2
Class 3 out-of-profile	AF12,AF13	1
Class 4 (best effort)	Default	0

A customer has commented that voice quality is not meeting expectations when calls cross your network. This voice packet was captured from the egress link to the customer. This capture is representative of the customer’s voice packets across your network and the queuing method shown is representative of packet treatment across your network. How might the problem be corrected?

A. Increase the bandwidth for the voice class.
B. Set up multiple priority queues to ensure low latency for voice packets.
C. Ensure that voice packets are correctly marked when they traverse your network.
D. Enable WRED in the CBWFQs to minimize the possibility of dropping voice packets.

Answer: C
Explanation:

QUESTION NO: 43

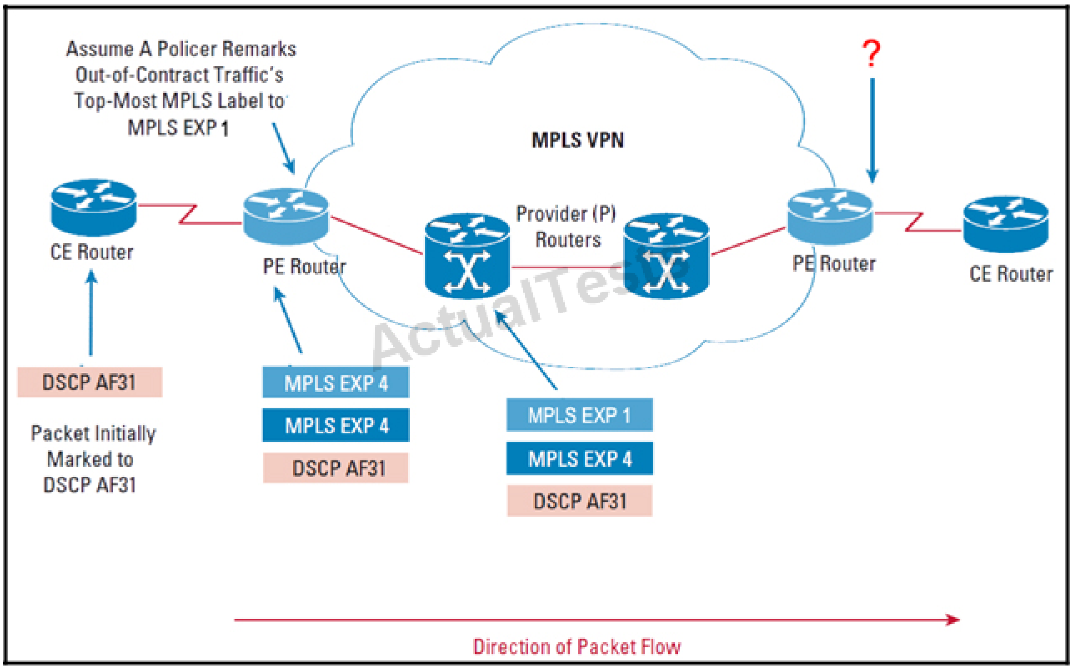
Which three can be monitored by IP SLA? (Choose three.)

A. jitter
B. packet loss
C. buffer allocation
D. network delay
E. queue utilization

Answer: A,B,D
Explanation:

QUESTION NO: 44

Refer to the exhibit.



You have implemented RFC 3270 Pipe mode in your MPLS network. A customer packet with the DSCP value of AF31 has been sent to the ingress PE router. The PE router has set the MPLS experimental bits per policy, but the policer has remarked the experimental bits of the top label. What will the DSCP value of the packet be when it leaves the egress PE router?

A. BE
B. CS1
C. CS3
D. CS4
E. AF31

Answer: E
Explanation:

QUESTION NO: 45

When the RFC 3270 Pipe mode is used, the PHB at the egress router is based on which of these?

A. The topmost MPLS label EXP value.
B. The MPLS VPN label EXP value.
C. The Differentiated Services field value in the IP packet header.
D. The packet is re-evaluated, remarked, and the PHB is based on this new marking.

Answer: A
Explanation:

QUESTION NO: 46

Which is the preferred queuing method provided by Cisco IOS Software for a converged VoIP network?

A. WFQ
B. CBWFQ
C. LLQ
D. FIFO

Answer: C
Explanation:

QUESTION NO: 47

Which Cisco IOS feature is used to enable the following class map configuration?

class-map match-all test-cm

match protocol gnutella file-transfer "*"

A. NetFlow
B. CBWFQ
C. NBAR
D. Flexible Packets Matching
E. CAR

Answer: C
Explanation:

QUESTION NO: 48

What can be used to provide read-only access to QoS configuration information and statistics from a network management station?

A. NetFlow
B. CISCO-CLASS-BASED-QOS-MIB
C. SYSLOG

D. NBAR
E. IP SLA

Answer: B
Explanation:

QUESTION NO: 49

Refer to the exhibit.

```
class-map match-any REALTIME
   match dscp ipv4 46
   match dscp ipv4 40
!
class-map match-all CRITICAL-A
   match dscp ipv4 48
!
class-map match-any BULK
   match any
!
class-map match-any CRITICAL-B
   match dscp ipv4 26
   match mpls experimental topmost 4
!
policy-map POLICY-A
  class REALTIME
    police rate percent 35
      conform-action transmit
       exceed-action drop
     priority
   class CRITICAL-A
     bandwidth percent 20
   class BULK
      random-detect exp 1 2500 packets
        7500 packets
       random-detect exp 0 1500 packets
        10500 packets
   class CRITICAL-B
     bandwidth percent 15
!
interface TenGigE0/0/1/1
  service-policy output POLICY-A
```

A service provider moved from a three-class service provider QoS model to a four-class service provider QoS model. A new class of service called CRITICAL-B was created to support traffic with a DSCP marking of AF31. A customer reports that the traffic in the CRITICAL-B class is having problems. What step should the service provider take to correct this issue?

A. The class map for CRITICAL-B should have a DSCP value of 28.
B. The max-reserved-bandwidth command must be used to allocate more bandwidth than allowed by default.

C. The policy map should be updated to move the BULK class to the bottom of the policy.
D. The CRITICAL-A class should change from match-all to match-any.

Answer: C
Explanation:

QUESTION NO: 50

The provider you work for supports a wide range of Cisco router models and Cisco IOS Software releases. You are in the process of moving to a SNMP-based tool to monitor QoS configurations and statistics. What are some of the challenges to this approach?

A. The SNMP MIB provides direct access into the QoS configuration parameters. Poor configuration could lead to changes that are difficult to track.
B. Some models of Cisco routers do not have full support for the class-based QoS MIB, which results in a number of elements that cannot be managed.
C. Older versions of Cisco IOS Software do not maintain index persistency, which results in significant changes in the information that is collected unless properly configured.
D. SNMP should be used in conjunction only with Cisco IP SLA to ensure that the information gathered is reported properly.

Answer: C
Explanation:

QUESTION NO: 51

Which statement about when you mix TCP and UDP traffic in the same class and overflow occurs is true?

A. UDP traffic suffers more, because TCP guarantees delivery.
B. TCP traffic is starved, because UDP never slows down when packets get dropped.
C. The overflow is not a problem, because both TCP and UDP packets get randomly dropped and traffic will back out.
D. Mixing TCP and UDP in the same class is not supported.

Answer: B
Explanation:

QUESTION NO: 52

A provider plans to use IP SLA to help diagnose future performance issues in their PE routers in the network. Which two should be considered by the provider? (Choose two.)

A. IP SLA requires a significant amount of resources to operate properly. Using IP SLA on the PE routers will only aggravate performance-related issues the network already encounters.
B. IP SLA is supported only on Cisco IOS edge routers, because it simulates traffic from the network endpoints.
C. IP SLA should be implemented on dedicated hardware to properly simulate network endpoints.
D. IP SLA should use a mix of source and target points.

Answer: C,D
Explanation:

QUESTION NO: 53

Refer to the exhibit.

```
class-map match-any REALTIME
  match dscp ipv4 46
  match dscp ipv4 34
  match mpls experimental topmost 5
!
class-map match-any CRITICAL-DATA
  match dscp ipv4 26
  match mpls experimental topmost 3
!
policy-map PE-OUT
   class REALTIME
      police rate percent 35
           conform-action transmit
           exceed-action drop
        priority
    class CRITICAL-DATA
        bandwidth percent 40
    class class-default
        random-detect exp 1 2500 packets 7500 packets
        random-detect exp 0 1500 packets 10500 packets
!
interface TenGigE0/0/1/1
   service-policy output PE-OUT
```


CLASS	DSCP	EXP
Reserved for Control Plan Traffic	Class Selector 7	7
Reserved for Control Plan Traffic	Class Selector 6	6
Class 1 (real-time traffic)	EF	5
Class 2 in-profile	AF31	4
Class 2 out-of-profile	AF32,AF33	3
Class 3 in-profile	AF11	2
Class 3 out-of-profile	AF12,AF13	1
Class 4 (best effort)	Default	0

A provider uses the recommended mappings for DSCP to MPLS EXP. Several sites connected to the same PE router report that critical data has been experiencing a high number of drops. The customer marks its CRITICAL data as AF31. What appears to be the problem?

A. The classification for CRITICAL-DATA should contain a match-all, not a match-any.
B. The provider is using the recommended mappings for DSCP to MPLS EXP, so the CLASS-MAP for the CRITICAL-DATA should contain EXP value of 4.
C. The recommended mapping for AF31 is to MPLS EXP 5, which is classified in the REALTIME class, not the CRITICAL DATA class.
D. The allocated bandwidth for CRITICAL-DATA should be increased to provide sufficient bandwidth to support the application.
E. If the data receiving drops is high priority, it should be marked with MPLS EXP value 7.

Answer: B
Explanation:

QUESTION NO: 54

What are three fundamental QoS mechanisms? (Choose three.)

A. congestion avoidance
B. undersubscription
C. congestion handling
D. policing
E. marking

Answer: A,D,E
Explanation:

QUESTION NO: 55

Which three issues are important challenges for monitoring service quality in the SP core network? (Choose three.)

A. Multiple flows compete for a limited amount of bandwidth.
B. Packets have to traverse many network devices and links that add up to the overall delay.
C. Traffic flows are continuously high.
D. Packets may have to be dropped when a link is congested.
E. Relatively few tools are available for collecting service quality data.

Answer: A,B,D

Explanation:

QUESTION NO: 56

Refer to the exhibit.

```
class-map match-any SP-REALTIME
 match dscp ipv4 46
 match dscp ipv4 40
 match mpls experimental topmost 5
!
policy-map PE-THREE-CLASS-SP-MODEL
  class SP-REALTIME
    police rate percent 35
      conform-action transmit
      exceed-action drop
    priority
```


Which two statements about the SP-REALTIME traffic class are true? (Choose two.)

A. Traffic is queued using LLQ.
B. Traffic is policed to 35 percent of the interface bandwidth.
C. The priority configuration command is missing the maximum guaranteed bandwidth rate.
D. The police configuration command is missing the violate-action option.
E. Traffic conforming to the police rate will be transmitted and all the markings on the packets will be cleared.

Answer: A,B

Explanation:

QUESTION NO: 57

Refer to the exhibit on a PE router. What could be causing the issue?

Service-policy output: OUTPUT-POLICY

Counters last updated 00:00:00 ago

Class-map: REALTIME (match-any)

845206 packets, 348763573 bytes

30 second offered rate 3000000 bps, drop rate 2500000 bps

Match: ip dscp cs5 (40) ef (46)

Priority: 10% (500 kbps), burst bytes 327265678, b/w exceed drops: 312925450

Class-map: DSCP-OUT-RTVI (match-any)

0 packets, 0 bytes

30 second offered rate 0000 bps, drop rate 0000 bps

Match: ip dscp af41 (0)

Queueing

queue limit 32 packets

(queue depth/total drops/no-buffer drops) 0/0/0

(pkts output/bytes output) 0/0

Class-map: class-default (match-any)

0 packets, 0 bytes

5 minute offered rate 0000 bps, drop rate 0000 bps

Match: any

Queueing

queue limit 1024 packets

(queue depth/total drops/no-buffer drops) 0/0/0

(pkts output/bytes output) 0/0

A. PE interface was not configured with enough bandwidth.
B. Traffic rate CE sent is over the SLA limit.
C. PE needs to upgrade to a higher throughput interface.
D. CE marked all traffic incorrectly to DSCP 46.
E. CE marked all traffic incorrectly to DSCP 0.

Answer: D
Explanation:

QUESTION NO: 58

A service provider recently changed from RFC 3270 Uniform mode to Pipe mode. Which three considerations about this change are true? (Choose three.)

A. If the provider moved to a Short Pipe mode model, it would allow them to implement customer-specific QoS policy on the PE egress.
B. The Pipe mode provides the isolation between the customer and provider DiffServ domains, while maintaining a simpler support model.
C. The Short Pipe mode derives the outbound classification of the WRED and WFQ based on the DiffServ policy of the provider.
D. In Uniform mode, when the topmost EXP value is changed, the changed EXP value will propagate to the original IP precedence or DSCP.

Answer: A,B,D
Explanation:

QUESTION NO: 59

What would be the first step when defining an MQC?

A. Add one or more match commands to classify packets.
B. Attach the traffic policy (policy map) to the interface by using the service-policy command.
C. Define a traffic class by using the class-map command.
D. Apply the commands such as bandwidth, police, and set dscp to enable the QoS feature.
E. Create a traffic policy by using the policy-map command.

Answer: C

Explanation:

QUESTION NO: 60

Refer to the exhibit.

```
Service-policy output: mypolicy

Class-map: voice (match-all)
0 packets, 0 bytes
5 minute offered rate 0 bps, drop rate 0 bps
Match: ip precedence 5
Weighted Fair Queueing
Strict Priority
Output Queue: Conversation 264
Bandwidth 128 (kbps) Burst 3200 (Bytes)
(pkts matched/bytes matched) 0/0
(total drops/bytes drops) 0/0

Class-map: gold (match-all)
0 packets, 0 bytes
5 minute offered rate 0 bps, drop rate 0 bps
Match: ip precedence 2
Weighted Fair Queueing
Output Queue: Conversation 265
Bandwidth 100 (kbps) Max Threshold 64 (packets)
(pkts matched/bytes matched) 0/0
(depth/total drops/no-buffer drops) 0/0/0

<output omitted>

Class-map: class-default (match-any)
0 packets, 0 bytes
5 minute offered rate 0 bps, drop rate 0 bps
Match: any
```


Which two statements are correct? (Choose two.)

A. The voice class is configured with the priority 128 command.
B. The voice class is configured with the bandwidth 128 command.
C. The gold class is configured with the priority 100 command.
D. The gold class is configured with the bandwidth 100 command.
E. The class-default class is configured with the fair-queue command.
F. The voice class is configured with the police 128 command.

Answer: A,D
Explanation:

QUESTION NO: 61

Refer to the exhibit.

```
Service-policy output:policy1

Class-map:class1 (match-all)
0 packets, 0 bytes
5 minute offered rate 0 bps, drop rate 0 bps
Match:none
Weighted Fair Queueing
Output Queue:Conversation 265
Bandwidth 50 (%)
Bandwidth 772 (kbps) Max Threshold 64 (packets)
(pkts matched/bytes matched) 0/0
(depth/total drops/no-buffer drops) 0/0/0

Class-map:class2 (match-all)
0 packets, 0 bytes
5 minute offered rate 0 bps, drop rate 0 bps
Match:none
Weighted Fair Queueing
Output Queue:Conversation 266
Bandwidth 25 (%)
Bandwidth 386 (kbps) Max Threshold 64 (packets)
(pkts matched/bytes matched) 0/0
(depth/total drops/no-buffer drops) 0/0/0
```


Which queueing method has been configured?

A. LLQ
B. CBWFQ
C. WFQ
D. MDRR
E. priority queueing

Answer: B

Explanation:

QUESTION NO: 62

Which two Cisco IOS commands can be used to monitor QoS on a device? (Choose two.)

A. show class-map
B. show qos-maps
C. show policy-map
D. show service-policy
E. show qos-interface

F. show qos-dbl

Answer: A,C
Explanation:

QUESTION NO: 63

A service provider will apply this policy at which of these interfaces?

```
interface Gi1/1

service-policy output SP-POLICY

!

policy-map SP-POLICY

class TRAFFIC-CLASS-1

police rate percent 20 conform-action transmit exceed-action drop

priority

!

class TRAFFIC-CLASS-2

bandwidth percent 50

queue-limit 800 ms

random-detect exp 3 280 ms 490 ms

!

class class-default

queue-limit 500 ms

random-detect exp 1 200 ms 350 ms

random-detect exp 0 100 ms 200 ms

!

end-policy-map

!
```

class-map match-any TRAFFIC-CLASS-1

match dscp ipv4 45 ef cs6

match mpls experimental topmost 5 6

end-class-map

!

class-map match-any TRAFFIC-CLASS-2

match dscp ipv4 cs2 af21 af23 af31 af32 af33 35 37

match mpls experimental topmost 3

end-class-map

!

A. CE interface toward PE
B. PE interface toward CE
C. PE interface toward P
D. P interface toward PE

Answer: C
Explanation:

QUESTION NO: 64

Which three match statements are valid in Cisco IOS XR Software? (Choose three.)

A. match field
B. match mpls experimental topmost
C. match vpls
D. match flow
E. match qos-group
F. match port-type

Answer: B,C,E
Explanation:

QUESTION NO: 65

Refer to the exhibit.

```
Router# show class-map

class-map match-all REALTIME
match dscp ef
match dscp cs4
!
class-map match-all VOIP
match dscp ef
match protocol rtp
!
class-map BULK-DATA
match dscp af11 af12 af13
```


The QoS profile was recently updated to mark VoIP traffic with a DSCP setting of AF41. However, the traffic going to the phones in the 172.19.5.0/24 address space is not being marked properly. What is causing this problem?

A. The traffic policy is not written correctly, which causes the wrong marking to be applied.
B. The VoIP classification should be configured before any other classifications, because it is the highest priority.
C. The access list does not properly identify SIP traffic.
D. Class typeA markings appear first in the policy map, which causes this marking to be applied.

Answer: A
Explanation:

QUESTION NO: 66

Refer to the exhibit.

```
Router# show class-map

class-map match-all REALTIME
match dscp ef
match dscp cs4
!
class-map match-all VOIP
match dscp ef
match protocol rtp
!
class-map BULK-DATA
match dscp af11 af12 af13
```


Which two statements are correct? (Choose two.)

A. No traffic can be matched to the REALTIME traffic class.
B. No traffic can be matched to the VoIP traffic class.
C. No traffic can be matched to the BULK-DATA traffic class.
D. AF11 or AF12 or AF13 marked packets will be matched to the BULK-DATA traffic class.
E. The BULK-DATA traffic class is missing the match-any option in the class-map command.

Answer: A,D
Explanation:

QUESTION NO: 67

Refer to the exhibit.

```
PE-SP#show policy-map interface fastEthernet 0/0

FastEthernet0/0
 Service-policy output: PE-THREE-CLASS-SP-MODEL
   queue stats for all priority classes:
     Queueing
     queue limit 64 packets
      (queue depth/total drops/no-buffer drops) 0/0/0
      (pkts output/bytes output) 133836/119447014
   Class-map: SP-REALTIME (match-any)
     133836 packets, 119447014 bytes
     5 minute offered rate 0000 bps, drop rate 0000 bps
     Match: dscp ef (46)
       13823 packets, 3342436 bytes
       5 minute rate 0 bps
     Match: dscp cs5 (40)
       120013 packets, 116104578 bytes 5 minute rate 0 bps
     Match: mpls experimental topmost 5
       0 packets, 0 bytes
       5 minute rate 0 bps
     police: rate 35 %
       (35000000 bps, burst 1093750 bytes)
      conformed 133836 packets, 119447014 bytes; action:
        transmit
      exceeded 0 packets, 0 bytes; action:
        drop
```


Which statement is correct?

A. This policy map is applied to the FastEthernet0/0 interface in the input direction.
B. No traffic matched the SP-REALTIME class.
C. A total of 133836 packets have been transmitted out of the FastEthernet0/0 interface.
D. Packets marked with both DSCP EF and MPLS EXP 5 markings will not be classified into the SP-REALTIME traffic class.
E. The FastEthernet0/0 interface is using CBWFQ egress queueing.

Answer: C
Explanation:

QUESTION NO: 68

You work for service provider that employs a four-class service provider QoS model (best effort, bronze, silver, gold). A customer recently reported an issue with high drop rate on a mission-critical database application in the silver class. The silver class is currently at capacity. The gold class is currently only used for voice and video conferencing, and has available bandwidth. The application was remapped to the gold class. Which statements are valid alternative options to this decision? (Choose two.)

A. Moving to a three-class QoS model would allow for more bandwidth per class and would reduce the support of maintaining multiple classes.

B. The provider should consider moving to a six-class QoS model to provide more available classes of service to differentiate traffic.
C. In addition to the remapping, the provider would have to change the queue model from WFQ to priority queued.
D. Additional bandwidth should have been added to the silver class from one of the lower classes before the traffic class was moved.

Answer: B,D
Explanation:

QUESTION NO: 69

Which three items are important when monitoring service quality in the SP network? (Choose three.)

A. The typical cause of QoS issues is intermittent failure of network devices and these are inherently difficult to detect.
B. QoS issues can be difficult to detect when there is not enough traffic to trigger the configured QoS policy.
C. Establishing alarm thresholds for QoS problems can be complicated.
D. QoS support differs from device to device, depending on underlying hardware platforms and software.
E. Collection of data to detect QoS issues is slow and labor-intensive.
F. Troubleshooting class map configurations is complex and frequently required.

Answer: B,C,D
Explanation:

QUESTION NO: 70

What are two benefits of the DiffServ model? (Choose two.)

A. Its class-based mechanisms are simple and scale well.
B. It provides absolute guarantees of service quality.
C. Service quality can be tailored to the needs of many different types of traffic.
D. It does not require setting up an end-to-end signaling mechanism.

Answer: C,D
Explanation:

QUESTION NO: 71

Which three components would you use to define and apply traffic classification? (Choose three.)

A. route map
B. service policy
C. class map
D. route policy
E. policy map

Answer: B,C,E
Explanation:

QUESTION NO: 72

Which Cisco IOS IP SLA Probe would you select to identify bottlenecks in the network?

A. ICMP Path Jitter
B. ICMP Echo
C. ICMP Path Echo
D. FTP Path Echo

Answer: C
Explanation:

QUESTION NO: 73

When the customer LAN is managed by the service provider and the network is oversubscribed, what is the best location for use of congestion avoidance mechanisms?

A. on output interfaces from the customer LAN leading into the service provider network
B. in interfaces in the service provider network incoming from the customer LAN
C. in the service provider core
D. on output interfaces from the service provider network leading to the customer LAN

Answer: A
Explanation:

QUESTION NO: 74

To classify and prioritize traffic according to importance, which three statements are correct? (Choose three.)

A. Each class should be guaranteed at least some bandwidth.
B. Voice traffic should be given top priority.
C. Email should be placed in a best-effort class.
D. Critical applications should be given sufficient bandwidth to meet their requirements.
E. Video should be placed in the same class as voice.

Answer: B,C,D
Explanation:

QUESTION NO: 75

Which tool is best suited to provide incident management or trouble ticketing?

A. Remedy
B. CiscoWorks
C. Tivoli
D. Voyence
E. Intelliden

Answer: A
Explanation:

QUESTION NO: 76

Where would a service provider apply this policy?

interface Gi1/1

service-policy output SP-POLICY

policy-map SP-POLICY

class CLASS-REALTIME

```
priority
set cos 5
!
class CLASS-HIPRIORITY
bandwidth remaining percent 75
set cos 3
!
class CLASS-BASIC
bandwidth remaining percent 20
set cos 1
!
class class-default
set cos 1
!
end-policy-map

class-map match-any CLASS-BASIC
match dscp af11 af12 af13
end-class-map
!
class-map match-any CLASS-REALTIME
match dscp ef af42 af43
end-class-map
!
class-map match-any CLASS-HIPRIORITY
match dscp af21 af22
end-class-map
```

!

A. CE interface toward PE
B. PE interface toward CE
C. PE interface toward P
D. P interface toward PE

Answer: B
Explanation:

QUESTION NO: 77

An MPLS VPN customer reports that the VoIP voice quality is very bad for calls across the MPLS VPN. What could be the problem?

A. Traffic shaping is applied to the voice traffic class.
B. The voice traffic class is using the priority command.
C. The voice packets are marked as DSCP EF.
D. WRED is not configured for the voice traffic class.
E. The voice traffic class is using the default queue size.

Answer: A
Explanation:

QUESTION NO: 78

A customer is experiencing poor quality on voice connections coming from a particular branch office. Data on the service provider network capacity indicates that the devices serving that location are heavily oversubscribed. It will be months before planned expansions can be implemented. Which class-based traffic-shaping technique would be appropriate to consider implementing?

A. shaping to the committed rate
B. shaping to the peak rate
C. shaping to the observed median rate
D. shaping to the configured average rate

Answer: D
Explanation:

QUESTION NO: 79

A customer with an unmanaged CE router has reported experiencing poor quality on the video conferencing and broadcast multimedia applications. To address the issue, the provider has implemented shaping on the PE router to "smooth out" speed mismatches in the network. Which of these is a valid concern with this approach?

A. In addition to shaping, the provider should implement policing so the application can detect the congestion and throttle traffic.
B. Shaping can only be applied to the outbound interface, which does not prevent the customer from flooding the inbound link to the provider.
C. Shaping should only be used in a TCP-based environment where the application can detect the congestion and adjust. Shaping provides little benefit with UDP applications.
D. While shaping may help the broadcast multimedia application, it could cause additional delay in the video conferencing application.

Answer: D
Explanation:

QUESTION NO: 80

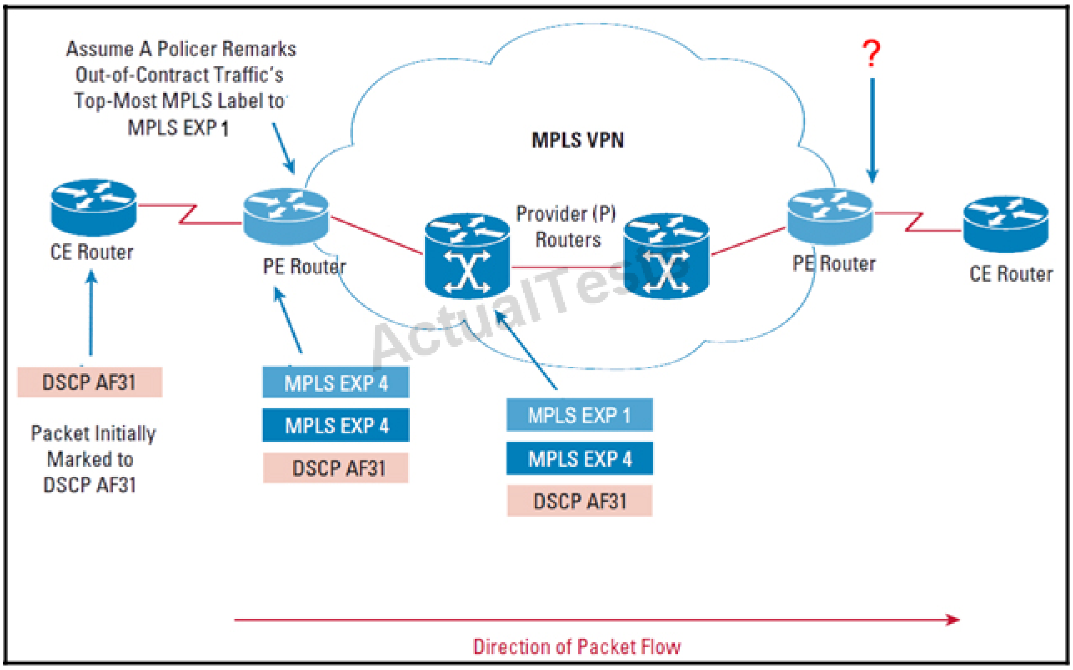
Which two match statements are correct in Cisco IOS Software? (Choose two.)

A. class-map class1
match not protocol ip
B. class-map match-none class2
match protocol ip
C. class-map class3
match dscp af31
match dscp af41
D. class-map match-any class3
match dscp af31
match dscp af41

Answer: A,D
Explanation:

QUESTION NO: 81

Refer to the exhibit.



You have implemented RFC 3270 Short Pipe mode in your MPLS network. A customer packet with the DSCP value of AF31 has been sent to the ingress PE router. The PE router has set the MPLS experimental bits per policy, but the policer has remarked the experimental bits of the top label. What will the DSCP value of the packet be when it leaves the egress PE router?

A. BE
B. CS1
C. CS3
D. CS4
E. AF31

Answer: E

Explanation:

QUESTION NO: 82

When the RFC 3270 Short Pipe mode is used, the PHB at the egress router is based on which of these?

A. The topmost MPLS label EXP value.
B. The MPLS VPN label EXP value.
C. The Differentiated Services Field value in the IP packet header.
D. The packet is re-evaluated, remarked, and the PHB is based on this new marking.

Answer: C

Explanation:

QUESTION NO: 83

When the RFC 3270 Uniform mode is used, the PHB at the egress router is based on which of these?

A. The topmost MPLS label EXP value.
B. The MPLS VPN label EXP value.
C. The Differentiated Services field value in the IP packet header.
D. The packet is re-evaluated, remarked, and the PHB is based on this new marking.

Answer: A

Explanation:

QUESTION NO: 84

Refer to the exhibit.

```
!
class-map match-all SP-REALTIME
   match dscp ipv4 46
   match dscp ipv4 36
   match mpls experimental topmost 5
!
class-map match-any SP-CRITICAL-DATA
    match dscp ipv4 48
    match dscp ipv4 34
    match dscp ipv4 26
    match mpls experimental topmost 3 6
!
policy-map PE-THREE-CLASS-SP-MODEL
  class SP-REALTIME
     police rate percent 35
       conform-action transmit
       exceed-action drop
     priority
  class SP-CRITICAL-DATA
     bandwidth percent 40
     random-detect exp 3 15000 packets 45000 packets
   class class-default
     random-detect exp 1 2500 packets 7500 packets
     random-detect exp 0 1500 packets 10500 packets
!
interface TenGigE0/0/1/1
service-policy output PE-THREE-CLASS-SP-MODEL
```

The service provider uses a three-class service provider QoS model (real-time, critical, best effort). The provider receives reports from a customer about poor quality on their multimedia conferences. The provider recently updated their profiles to include multimedia conferencing (af41) to their real-time class, which includes voice (ef). What are the two most likely causes of this problem? (Choose two.)

A. The class map for SP-REALTIME does not include the DSCP values for multimedia conferences and voice traffic.
B. The MPLS experimental 3 is in the SP-CRITICAL-DATA class, not the SP_REALTIME class.
C. Multimedia conferencing is included in the SP-CRITICAL-DATA class.
D. The SP-REALTIME class should include mpls experimental 4, not 5.
E. The SP-CRITICAL-DATA class should not use random detect for multimedia traffic.

Answer: A,C

Explanation:

QUESTION NO: 85

Refer to the exhibit.

```
access-list 100 permit udp any any range 16384 32767
access-list 101 permit tcp any any eq 1720
!
class-map match-all voip
match ip precedence 5
match ip dscp cs7
match access-group 100
!
class-map match-all control
match ip precedence 5
match ip dscp cs7
match access-group 101
!
policy-map test
class voip
priority 512
class control
bandwidth 256
```

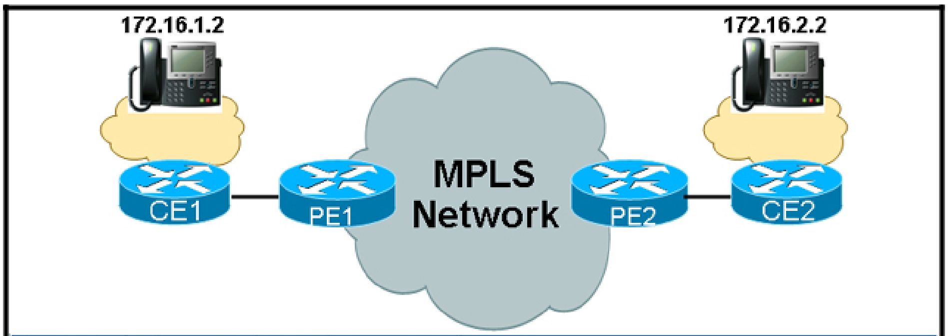

The voice traffic is not receiving the appropriate per-hop behavior at the egress of the CE-PE link. What is the problem?

A. Both ACLs are misconfigured.
B. Both class maps are misconfigured.
C. The control traffic class should be configured with the priority command instead of the bandwidth command.
D. Instead of hard coding the actual bandwidth on the priority and bandwidth commands, the percent option should be used.

Answer: B
Explanation:

QUESTION NO: 86

Refer to the exhibit.




```
⊞ Frame 42 (78 bytes on wire, 78 bytes captured)
⊟ Ethernet II, Src: Cisco_ff:a0:c1 (00:07:0e:ff:a0:c1), Dst: Cisco_d5:b9:80 (00:07:0e:d5:b9:80)
  ⊞ Destination: Cisco_d5:b9:80 (00:07:0e:d5:b9:80)
  ⊞ Source: Cisco_ff:a0:c1 (00:07:0e:ff:a0:c1)
    Type: IP (0x0800)
⊟ Internet Protocol, Src: 172.16.1.2 (172.16.1.2), Dst: 172.16.2.2 (172.16.2.2)
    Version: 4
    Header length: 20 bytes
  ⊞ Differentiated Services Field: 0xa0 (DSCP 0x28)
    Total Length: 64
    Identification: 0x2020 (8224)
  ⊞ Flags: 0x00
    Fragment offset: 0
    Time to live: 126
    Protocol: UDP (0x11)
  ⊞ Header checksum: 0xc0c8 [correct]
    Source: 172.16.1.2 (172.16.1.2)
    Destination: 172.16.2.2 (172.16.2.2)
⊟ User Datagram Protocol, Src Port: 49594 (49594), Dst Port: 49606 (49606)
    Source port: 49594 (49594)
    Destination port: 49606 (49606)
    Length: 44
  ⊞ Checksum: 0x13b6 [validation disabled]
⊞ Data (36 bytes)

0000  00 07 0e d5 b9 80 00 07  0e ff a0 c1 08 00 45 a0   ........ ......E.
0010  00 40 20 20 00 00 7e 11  c0 c8 ac 10 01 02 ac 10   .@  ..~. ........
0020  02 02 c1 ba c1 c6 00 2c  13 b6 80 04 b6 a5 00 09   .......,  ........
0030  b0 28 68 74 94 d4 70 b3  7e d5 aa 75 d4 28 79 92   .(ht..p. ~..u.(y.
0040  87 c3 ce 9f a7 f1 42 a9  59 15 30 8c 76 bb         ......B. Y.0.v.
```

```
PE2#show policy-map interface ethernet0/0/1
 Ethernet0/0/1

 Service-policy output: voice_priority

   Class-map: voice (match-all)
     0 packets, 0 bytes
     5 minute offered rate 0 bps, drop rate 0 bps
     Match: ip dscp ef
     Queueing
       Strict Priority
       Output Queue: Conversation 264
       Bandwidth 320 (kbps) Burst 8000 (Bytes)
       (pkts matched/bytes matched) 0/0
       (total drops/bytes drops) 0/0

   Class-map: video (match-all)
     0 packets, 0 bytes
     5 minute offered rate 0 bps, drop rate 0 bps
     Match: ip dscp af41
     Queueing
       Output Queue: Conversation 265
       Bandwidth 50 (%) Max Threshold 64 (packets)
       (pkts matched/bytes matched) 0/0
       (depth/total drops/no-buffer drops) 0/0/0

   Class-map: critical (match-all)
     6647 packets, 757758 bytes
     5 minute offered rate 21000 bps, drop rate 0 bps
     Match: ip dscp cs3
     Queueing
       Output Queue: Conversation 266
       Bandwidth 35 (%) Max Threshold 64 (packets)
       (pkts matched/bytes matched) 1/114
       (depth/total drops/no-buffer drops) 0/0/0

   Class-map: class-default (match-any)
     10729 packets, 1144263 bytes
     5 minute offered rate 6000 bps, drop rate 0 bps
     Match: any
     Queueing
       Output Queue: Conversation 267
       Bandwidth 15 (%) Max Threshold 64 (packets)
       (pkts matched/bytes matched) 15/4085
       (depth/total drops/no-buffer drops) 0/0/0
PE2#
```

A customer reports that data between their sites across your network is very slow and that voice and video quality are unacceptable. Packet traces of data coming in from and going out to the customer confirm that many packets are being dropped in your network, and many of those that are not dropped are being remarked to CS1. Class maps and a policy map configured as shown are applied inbound to the Fast Ethernet interfaces that connect the customer. What could be the problem?

A. The CIR and PIR values appear to have been configured using kbps when the entries are bps.
B. The match criteria is DSCP and the customer packets are being sent marked using IP precedence.
C. All traffic is matching the class class-default, because all other classes are match-all with multiple match criteria.
D. The customer data is being rate limited, but is not being queued by class, which is causing tail drops and packet remarking.

Answer: A
Explanation:

QUESTION NO: 87

Refer to the exhibit.

```
PE1# show policy-map interface GigabitEthernet1/0

GigabitEthernet1/0

Service-policy input: set-MPLS
Class-map: GOLD (match-all)
        652524 packets, 554645400 bytes
        5 minute offered rate 16450 bps, drop rate 0 bps
        Match: ip precedence 5
        police:
              cir 1000000 bps, bc 31250 bytes
           conformed 652524 packets, 0 bytes; actions:
              set-mpls-exp-imposition-transmit 5
           exceeded 0 packets, 0 bytes; actions:
              drop
           conformed 0 bps, exceed 0 bps
```

Which statement is correct?

A. This is a dual-rate policing example.
B. Packets marked with IP Precedence 5 that conforms to the committed rate will be marked with MPLS EXP 5 then transmitted.
C. Packets marked with IP Precedence 5 that exceeds the committed rate will be marked with MPLS EXP 5.
D. This policy will police traffic at the ingress and egress directions.

Answer: B
Explanation:

QUESTION NO: 88

A customer reports that during peak hours, FTP performance is unacceptable. Currently, the service provider maps the customer FTP traffic into the class-default traffic class and FTP traffic is marked with MPLS EXP1. The class-default class traffic is configured with MPLS EXP based WRED and also with the bandwidth command. What could be causing the FTP performance issues with the customer?

A. The FTP-type bulk traffic should be mapped to MPLS EXP 5 instead of MPLS EXP 1. The improper marking of FTP packets is causing CBWFQ to assign lower priority for FTP packets within the class-default traffic class.
B. The service provider is using LLQ on all the core links, but the FTP traffic is not is assigned to the priority queue.
C. The bandwidth command is causing the FTP issues. It is generally not recommended to use CBWFQ for the class-default traffic class.
D. FTP packets are being more randomly dropped that the other TCP traffic within the class-default traffic class, because the WRED thresholds for MPLS EXP 3 marked packets were configured to improper values.

Answer: D
Explanation:

QUESTION NO: 89

Refer to the exhibit.

```
Router# show policy-map policy1

Policy Map policy1
 Class class1
  police cir percent 20 bc 300 ms pir percent 40 be 400 ms
   conform-action transmit
   exceed-action drop
   violate-action drop
```

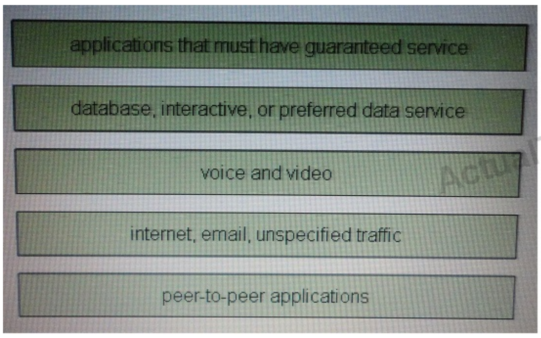
Which statement is correct?

A. This example is a dual-bucket policer, and only traffic that exceeds the committed rate will be dropped.
B. This example is a dual-rate policer, and traffic exceeding either the peak or the committed rate will be dropped.
C. This example is a traffic policer, and three token buckets are used to meter the incoming traffic.
D. This example is a traffic policer, and traffic exceeding the committed rate will be buffered until the congestion is cleared.
E. This example is a percentage-based policer, and the configured percentage is referenced to the input or output queue size.

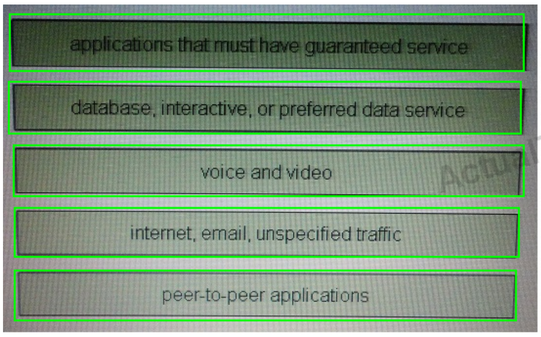
Answer: B
Explanation:

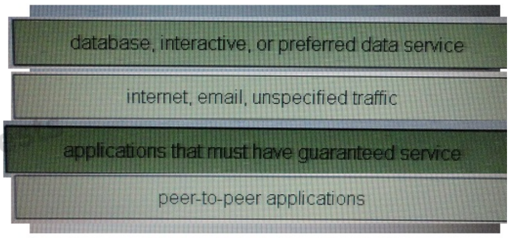
QUESTION NO: 90 DRAG DROP

Drag the definitions from the left onto the traffic classification on the right. Not all definitions will be used.



Answer:

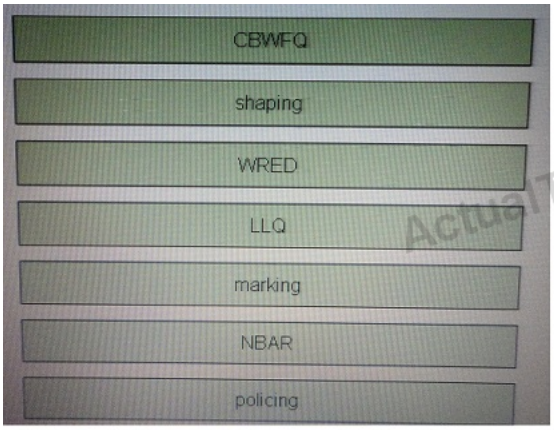


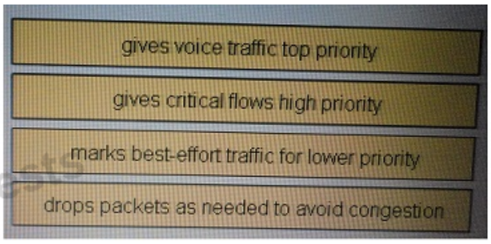


Explanation: Application that must have guaranteed service mission-critical
Database, interactive, or preferred service transactional
Internet, email, unspecified traffic best-effort
Peer-to-peer applications scavenger

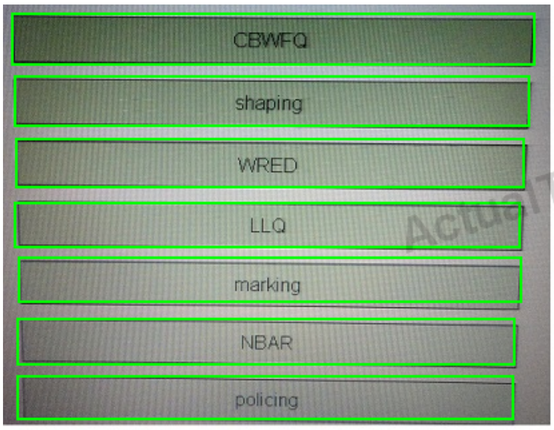
QUESTION NO: 91 DRAG DROP

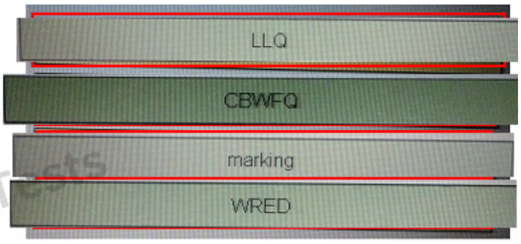
Drag and drop mechanisms on the left to their corresponding uses on the right. Some mechanisms may not used.





Answer:





Explanation:

CBWFQ gives critical flows top priority

WRED drops packets as needed to avoid congestion

LLQ gives voice traffic top priority

Marking marks best effort traffic for lower priority